

GROUP NUMBER: 35

GROUP MEMBERS: Francesco Pizzato

AVAILABLE INPUTS: Input files are available in the hdfs file system: /data/BDC2526

TEST: The goal of this test is **to assess the scalability of your MRFairFFT program**. The test must be performed on the artificial datasets ArtificialNx DyKz.csv found on the hdfs file system. Conventionally, the name of the dataset represents some basic information: ArtificialNx DyKz.txt contains x points in R^y forming z natural clusters.

Use this table to report the results of the requested experiments. For all experiments use $kA=8$, $kB=2$, $L=16$, and filling only those cells for which the execution required at most 5 minutes.

Dataset	Runtime (2 executors)	Runtime (4 executors)	Runtime (8 executors)	Runtime (16 executors)	Obj. function (16 executors)
ArtificialN130kD7k10	614ms	407ms	354ms	265ms	38.98211736375455
ArtificialN1.3mD7k10	2372ms	2013ms	1792ms	1561ms	39.571839080279496
ArtificialN13mD7k10	28474ms	13707ms	12122ms	10357ms	39.697057789490074
ArtificialN130mD7k10	321851ms	167490ms	79080ms	46844ms	39.86808928745398

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? Yes
- (Only if you answered YES to previous question) Describe in at most 10 lines how you used the Generative AI tool.

I used Claude AI to set up the working environment, solving some compatibility issues between Spark, Windows, and IntelliJ during configuration. Then, I used it to check some sections of my code, and it suggested using a double[] minDist array to store the minimum distance of each point from the current set of centers. This helped me avoid redundant recomputations at every iteration and simplified the overall logic of the FairFFT procedure. Finally, I used it to help me connect to the cluster and go through the ssh procedure, which had caused me some issues.